

SLIDE 1 | Problem and Proposed Solution

SIH 26175 • Single-View Height Estimation and 3D Flythrough • ISRO / Space Applications Centre • Disaster Management

The problem

A single nadir satellite image records colour, not height. From directly overhead a ten-storey building and a painted rectangle look nearly identical. Yet elevation is what disaster response needs: which streets are low-lying, how a slope has changed, which routes remain passable, which structures have collapsed.

Conventional sources are too slow. Airborne LiDAR needs an aircraft, crew and a scheduled flight. Stereo pairs require the satellite to be tasked for two viewing angles. InSAR needs specialised radar satellites and slow interferometric processing. All take days to weeks. Disaster response needs hours, from imagery that already exists.

Our solution

DepthWizard converts a single optical satellite image into a metric Digital Surface Model and a navigable LOD1 3D city model, running entirely on CPU with no GPU and no network connection required at inference time.

What makes it different

Every comparable submission feeds the image to a monocular depth foundation model. **We implemented that, measured it, and our system rejects it automatically.** Depth Anything V2 calibrated against reference elevation at $R^2 = 0.099$, below our acceptance threshold.

Near-orthographic viewing geometry removes the perspective cues those models depend on. This is stated in the problem statement's own background and confirmed in the literature:

"Near-orthographic viewing geometry eliminates perspective height cues, forcing reliance on shadow information for height inference." — arXiv:2604.02009

Instead we use **shadow photogrammetry**: $h = L \times \tan(\theta_{\text{sun}})$. Measured geometry, not inference. Buildings are reconstructed as flat-topped extruded prisms (LOD1), not a smoothed heightfield — a heightfield interpolates between grid cells and is mathematically incapable of producing a vertical wall.

Four self-diagnosing gates mean the system refuses to claim metric heights it cannot justify: no georeferencing produces a dimensionless rDSM; incoherent shadows are downgraded to relative; composite mosaics are detected and declined; depth calibration below threshold is rejected. Coverage is reported in three honest categories — measured, inferred, failed — never as a single 100 percent figure.

SLIDE 2 | Technical Approach

Pipeline, height cues, solar geometry resolution and stack

Pipeline

single-acquisition image with solar metadata → GeoTIFF detection (CRS, affine, GSD) → SAM instance segmentation with watershed splitting → shape-based classification (building / tree / road / ground) → shadow detection and matching → $h = L \times \tan(\theta)$ → multi-cue fusion → quality gates → LOD1 extrusion with source imagery draped → GeoTIFF export with sidecar JSON.

Five height cues, fused by inverse-variance weighting

Cue	Principle	Role
Shadow geometry	$h = L \times \tan(\theta)$, photogrammetry	Primary — metric
Facade visibility	Visible wall extent, off-nadir imagery	Secondary
Relief displacement	Roof offset from footprint $\propto$ height	Secondary
Depth-calibrated	DAv2 ordering scaled to shadow anchors	Fallback
Neighbourhood prior	Median of confident neighbours	Last resort

Uncertainty is floored at 0.5 m before weighting to avoid division by near-zero. Methods disagreeing beyond combined uncertainty flag the building low-confidence rather than averaging silently. Every building reports height, uncertainty, contributing methods, and whether it was directly measured or inferred.

Solar geometry priority chain

Priority	Source	Confidence
1	Product metadata (.IMD / .XML / .MTL / EXIF sidecar)	Exact
2	Computed from acquisition datetime + scene lat/lon (pysolar)	Exact orbital geometry
3	Azimuth measured from the image's own shadow offsets	Validation cross-check
4	User-supplied slider	Estimated, labelled as such

Correctness rule: when measured shadows win, they supply only the azimuth. Elevation is inherited from metadata, computed or user values — never a default. An earlier bug used a hardcoded 45-degree default here, understating every height by a factor of 3.49 at 74-degree sun.

Stack

Segment Anything (ViT-B) • Depth Anything V2 Small • rasterio / GDAL • pysolar • scikit-learn (RANSAC, Huber) • scikit-image, OpenCV • FastAPI with SSE progress • Three.js WebGL rasterisation • Docker.

Deliberate decision: no neural rendering. NeRF and Gaussian splatting were evaluated and rejected — the problem statement names raster engines, and rasterisation meets the requirement at a fraction of the compute cost.

SLIDE 3 | Validation Results

Antakya, Turkey • WorldView pre-earthquake • GSD 0.305 m • sun elevation 28.3° • Vantor Open Data, CC-BY 4.0

Per-crop results under the two-tier admission gate

Crop	R (weighted)	Azimuth Δ	Rayleigh p	Tier	Buildings
crop1	0.490	5.4°	0.0017	BORDERLINE	12 borderline, 0 failed
crop2	0.681	10.6°	—	MEASURED	22 of 23, 0 failed
crop3	0.275	40.6°	0.062	REJECTED	19 relative, 2 failed
crop4	0.384	48.7°	0.004	REJECTED	29 relative, 1 failed

Headline numbers

- **88 buildings across 4 crops** — 34 measured or borderline, 48 relative, 2 failed
- **crop2: 22 of 23 buildings directly measured from shadow geometry**, mean 4.6 m, median 2.8 m, range 1.8–14.7 m
- crop1: mean 4.3 m, median 2.4 m — independently consistent with crop2 on the same old-city fabric
- **Measured sun azimuth 172.6° vs computed 162.0° — agreement within 10.6°** between two entirely independent methods
- Heights consistent with Antakya's known 1–2 storey Ottoman stone housing

Operating envelope — derived, then validated

$$h_{\min} = \min_offset_px \times GSD \times \tan(\theta_{\text{sun}})$$

Site	Sun elevation	GSD	h min	Outcome
Antakya	28.3°	0.305 m	0.82 m	Method validated, heights credible
Kathmandu	74.0°	0.500 m	28 m	Most buildings below floor; 2–112 m implausible

Derived from first principles, then correctly predicted which acquisition would succeed and which would fail. This quantifies the operating envelope of single-view shadow photogrammetry as a function of solar geometry and ground sample distance.

The rejected case matters most

crop4 is statistically significant by Rayleigh test ($p = 0.004$) — 29 shadow pairs pointing consistently — yet its azimuth delta of 48.7° blocks admission. A statistical test alone would have accepted a systematic artefact. Significance proves the pattern is not random; physical agreement proves it is the sun. **Both gates are required, and we have a documented case where each alone would have failed.**

SLIDE 4 | Engineering Results and Limitations

Performance, model benchmarking and honestly stated gaps

Performance

Configuration	Detection	Runtime	Use
SAM 384 / 8 points	Lower recall	23.8 s	Speed-optimised
SAM 512 / 12 points	96% centroid recall	65 s	Default — accuracy-optimised

A deliberate accuracy-for-speed trade. Stage optimisation reduced the pipeline from an initial 536 s: shadow matching vectorised with a single pre-dilated label map instead of per-building dilations, SAM prompt grid tuned, tile inference batched, shape features vectorised with `scipy.ndimage`, SLIC iterations capped, and blocking inference moved to a worker thread so progress streams concurrently. **CPU only** — approximately 4 GB RAM, no GPU. Demo scenes are precomputed and render in under two seconds.

Model comparison

	MiDaS Small	Depth Anything V2 Small	Our shadow method
Weights	82 MB	99 MB	None
Output	Relative disparity	Relative disparity	Metric metres
Brightness/height corr.	0.17	0.64	Physics-derived
Building/road separation	8.5 m	55.8 m	Per-building measured
Calibration R^2	—	0.099 — rejected	n/a
Interpretable	No	No	Yes

Both depth models improve *relative ordering*; neither produces metric height. Only geometry does.

Limitations — stated, not hidden

- **Footprint boundary precision.** 96% centroid recall but 9.4% mask IoU: we locate nearly every building, but boundaries are imprecise in dense fabric. This is a mask quality problem, not a detection problem. Polygon regularisation is the identified fix.
- **Systematic height underestimate.** Median 3.4 m against an expected 3–3.5 m per storey. Root cause identified: shadow truncation. Shadows falling on neighbouring rooftops are masked out, so measured shadow length is shorter than true length.
- **No RMSE against ground truth yet.** No reference DSM existed for the validation sites. Reported results are internal consistency and physical plausibility, not measured error. The GAMUS evaluation harness is built with five passing sanity tests and awaits paired tiles.
- **OSM validation attempted and failed:** of 324 OSM buildings across four crops, **zero** carried a height tag and only 8 carried building:levels. Turkish cities are well mapped for footprints but not height.
- **Crops selected for building density**, so coverage is not scene-wide. Landscape breadth beyond urban is untested.

For context, the most rigorous competing submission reports MAE 4.40 m, RMSE 5.86 m and $R^2 = 0.23$ on a 32-tile GAMUS baseline, describing it in their own release notes as “honestly poor”. Several others claim “metric-level accuracy” with no published numbers.

SLIDE 5 | Feasibility, Scalability and Impact

Deployment readiness, scaling path and operational value

Feasibility

- Working end-to-end pipeline today — standard workstation, ~4 GB RAM, no GPU
- Entirely open-source and permissively licensed — **zero licence cost**
- ISRO products (Cartosat, Resourcesat) already carry the solar metadata the method requires
- Output is standard GeoTIFF (float32, source CRS, EGM96 datum) — drops straight into QGIS or ArcGIS with no workflow change
- Containerised with Docker, health check and precomputed demo scenes; deployable free on Hugging Face Spaces CPU Basic
- Offline after installation — which matters when disaster disrupts connectivity

Challenges and mitigations

Challenge	Mitigation
Shadows occluded in dense low-rise rows	Multi-cue fusion; honest coverage reporting rather than fabricated fill
Sun elevation error is multiplicative ($\tan \theta$)	Priority chain prefers metadata and computed values; uncertainty propagated into every error bar
Overcast or near-overhead sun	Feasibility pre-check warns before processing; system declares non-metric output
Composite mosaics have no single sun	Coherence gate detects and reports it distinctly from 'no shadows found'

Scalability

- **No region-specific training.** Shadow geometry is universal physics and does not degrade across geographies, unlike a model fine-tuned on one city's morphology
- Tiling with cosine-feathered blending handles arbitrary scene size at bounded memory
- Per-tile inference is embarrassingly parallel; scales linearly with cores
- Cloud Optimized GeoTIFF support streams windows from object storage without downloading whole scenes
- Marginal cost per scene is CPU time alone, making a national building-height layer from archive Cartosat imagery tractable

Impact

Disaster response. The flood module computes per building the submerged depth (clamped between zero and building height), freeboard, and status — dry, partially inundated or fully submerged — with affected-building counts and inundated area exportable to CSV for response planning. Pre- and post-event height differencing supports automated damage assessment.

Operational. Hours instead of weeks. No aircraft, no tasked stereo acquisition, no LiDAR survey. Works on archive imagery that already exists, offline, on field hardware.

Governance. Urban planning, building-stock inventory, informal settlement mapping. The self-diagnosing gates mean an operator cannot unknowingly act on unreliable output — critical when decisions affect safety.

SLIDE 6 | Research and References

Literature basis, datasets and anticipated jury questions

Shadow-based height estimation

- Liasis & Stavrou (2016). *Satellite images analysis for shadow detection and building height estimation*. ISPRS Journal of Photogrammetry and Remote Sensing, 119:437–450. [sciencedirect.com/science/article/abs/pii/S0924271616301939](https://www.sciencedirect.com/science/article/abs/pii/S0924271616301939)
- Qureshi et al. (2024). *Building Height Estimation Using Shadow Length in Satellite Imagery*. arXiv:2411.09411 — arxiv.org/abs/2411.09411
- Byrnside (2022). *Shadow-Based Automatic Building Height Estimation From High Spatial Resolution Satellite Imagery*. Missouri State University. bearworks.missouristate.edu/theses/3768/
- Hao et al. (2021). *Building height estimation via satellite metadata and shadow instance detection*.

Monocular depth and its limits on nadir imagery

- Yang et al. (2024). *Depth Anything V2*. arXiv:2406.09414
- *Test-Time Adaptation for Height Completion via Self-Supervised ViT Features and Monocular Foundation Models*. arXiv:2604.02009 — near-orthographic geometry eliminates perspective cues and forces reliance on shadows; illumination variation can cause height errors exceeding 30 percent
- *Depth2Elevation: Scale Modulation with Depth Anything Model for Single-view Remote Sensing Image Height Estimation*. TechRxiv — up to 42 percent relative improvement on GAMUS

Datasets and benchmarks

- ISPRS 2D Semantic Labeling — Potsdam. 38 patches, 5 cm GSD, orthophoto + DSM + normalised DSM. [isprs.org \(Open Resources > Benchmarks\)](https://isprs.org/OpenResources/Benchmarks)
- GAMUS aerial monocular height benchmark — huggingface.co/datasets/earthflow/GAMUS
- Vantor / Maxar Open Data Program — free 30–50 cm single-acquisition imagery, CC-BY 4.0. registry.opendata.aws/maxar-open-data/ • browser: opengeos.org/maplibre-gl-vantor
- SAC reference material for SIH 26175 — github.com/IMG-PROCESS-SAC/SIH2026/

Reported accuracy for shadow methods in the literature

±4.66 m global accuracy against LiDAR reference (Warsaw, several hundred thousand objects); 4.08 m mean absolute error (Shanghai, 15,966 buildings, ICESat-2 calibrated). A systematic finding across studies is that short buildings are over-estimated and tall buildings under-estimated — we therefore report error stratified by height band rather than a single aggregate.

Anticipated jury questions

Question	Answer
Why not a depth model like everyone else?	We built it and measured its calibration at $R^2 = 0.099$. The system rejects it automatically. Nadir geometry removes the cues those models need.
What is your dominant error source?	Sun elevation. Height scales with $\tan(\theta)$, so the error is multiplicative and affects every building equally. We propagate its uncertainty into every error bar.
Why is your mask IoU only 9.4 percent?	Centroid recall is 96 percent — we locate nearly every building. The gap is boundary precision in dense fabric, a mask quality issue. Polygon regularisation is the fix.
Why is the median height low?	Shadow truncation: shadows falling on neighbouring rooftops are masked out, shortening measured shadow length. Identified and documented, not yet corrected.
What is your vertical datum?	EGM96, declared explicitly in the exported GeoTIFF.
Where does it fail?	Dense low-rise settlements, overcast scenes, near-overhead sun, and composite mosaics. All four are detected and reported.